

Checkpoint 1

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1 Introduction

Following both the “VR desktop of the future” and “gaming of the future” themes, creating an immersive VR environment while incorporating organizational applications sounds like a perfect mix of each theme. This project aims to create a VR environment with two main features. The ability to create tasks inside the environment and organize them on a virtual calendar or board. Traditional VR interfaces rely on virtual keyboards as the primary method to enter text into VR. While simple, virtual keyboards can be cumbersome and difficult to effectively enter text in VR. Mid-air handwriting using a VR pen potentially offers a better solution as physically holding a pen while writing is a more natural motion compared to using a virtual keyboard. While prior work has evaluated virtual keyboards and handwriting techniques independently, fewer studies have directly compared these interaction techniques within a VR text creation aspect. This project plans to evaluate two different input techniques for entering text in a VR environment: mid-air handwriting using a VR pen and virtual keyboard typing.

2 Related Work

The article “Controller-based Text-input Techniques for Virtual Reality: An Empirical Comparison” compares four different types of virtual keyboards. They focused their experiment primarily on “text-entry rate and accuracy, usability, and user experience”(Boletsis, Kongsvik, 2019). They found that drum-like keyboard and ray casting stood out as the most prominent virtual keyboard options. Secondly, the article “Evaluating the Performance of Hand-Based Probabilistic Text Input Methods on a Mid-Air Virtual Qwerty Keyboard” determined that mid-air virtual keyboards benefited from the familiarity of real keyboards, but due to “lack of passive haptic feedback, unconstrained user motion, and potential tracking inaccuracies or observability issues encountered in this interaction setting typically degrades the accuracy of user articulations.”(Dudley et al, 2023). During their experiment they found the average user could get around 20-30 wpm and users with greater VR experience achieved peak entry rates of 40-45 wpm (Dudley et al, 2023). For mid-air handwriting, the article “Handwriting for Efficient Text Entry in Industrial VR Applications: Influence of Board Orientation and Sensory Feedback on Performance” found that writing on virtual surfaces has “high entry speed, usability with little training compared to other controller-based virtual text entry techniques.”(Fourrier et al, 2023) Although, the average WPM of the tested users was only around 14.5 WPM with a peak of 21.11 WPM.

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